Stage 2 Implementation Report

AI-powered Study Planner | EECS 3311 | Rafat Kabir
Progress draft: F01–F06 and UC-11, November 6, 2026.

1. Overview and status

Students can create courses, upload materials, generate summaries, extract concepts, ask questions about their materials, and generate personalized study schedules through a Vaadin GUI or Java CLI. Spring Boot stores courses, materials, summaries, concept sets, chat history, profiles, plans, and calendar events in H2. F03 uses local Ollama; the HTTP adapter also supports hosted OpenAI-compatible providers.

F01–F06 and UC-11 are implemented. F10 has event storage and plan synchronization; its full interface remains pending. F07–F09 and course editing/deletion remain pending. Authentication supports one locally configured account and does not count as a project feature.

2. Feature-to-implementation mapping

F01: Course Creation. Deterministic. References: Stage 1 Task 1.2 F01, UC-01, SD01, and Task 4 F01. Students enter a name, description, start date, and end date. The system validates the input, saves an owned course, and displays it on the dashboard. Video: pending.

Part	Files, responsibilities, and key methods
GUI	`frontend/.../views/MainView.java`: `showSignIn`, `showDashboard`, `openCourseForm`, `refreshCourses`, and `renderCourses` handle sign-in, forms, errors, and course cards. `themes/study-planner/styles.css` styles the UI.
API client	`frontend/.../course/CourseApi.java`: `createCourse`/`viewCourses` call the backend. `CourseApiFactory.connect` creates a client for each signed-in view. Frontend `Course` and `CreateCourseRequest` records carry API data.
REST API	`backend/.../course/CourseController.java`: `createCourse`, `viewCourses`, and `getCourse` expose POST/GET `/api/courses` and GET `/api/courses/{id}`.
Application and domain	`CourseService` coordinates creation and retrieval. `User.createCourse`/`viewCourses` create and retrieve owned courses. Backend `Course` validates domain data; `CreateCourseRequest` validates API fields.

Storage	`CourseRepository` defines storage operations. `JdbcCourseRepository.save/findByOwner/findByIdAndOwner` use SQL. `schema.sql` defines the table. H2 files persist under `backend/data/`.
Authentication and errors	`SecurityConfiguration` verifies credentials; `CsrfController.token` supports protected writes. `ApiExceptionHandler` handles invalid requests and storage failures. The authenticated student determines course ownership.
CLI	`cli/StudyPlannerCli.java`: `main/run` provide `create`, `list`, and `help` using the same authenticated API.

Flow: form or CLI → CourseController → CourseService → User.createCourse → CourseRepository → H2. The GUI then reloads the saved course list. Failed submissions show errors and preserve form input.

All four fields are required. Names allow 120 characters; descriptions allow 2000. Text is trimmed, and end dates must be on or after start dates. Same-day courses and duplicate names are allowed. Course queries are restricted to their owner.

F02: Upload Learning Materials. References: Stage 1 Task 1.2 F02, UC-02, SD01, and Task 4 F02. Open an owned course, select files, and view the saved materials and extracted text. Upload/validation/text extraction are deterministic; the planned AI processing belongs to later features. Video: pending.

Part	Implementation and key methods
GUI / client	`MainView.openCourse` opens `CourseMaterialsPanel`; upload, `refreshMaterials`, and `viewText` show results and errors. Each file offers View text and Download original. `CourseApi.uploadMaterial/listMaterials/materialContent/downloadMaterial` call the authenticated API.
API / service	`CourseController.uploadMaterial/listMaterials/materialContent/downloadMaterial` delegate to `MaterialService`, which verifies course ownership before access.
Domain / parsing	`Course.addMaterial` creates `LearningMaterial`; `getContent` exposes extracted text. `MaterialParser.parse` checks filenames and parses PDF, DOCX, or UTF-8 TXT using PDFBox, POI, or UTF-8 decoding.
Storage / CLI	`MaterialRepository` and `JdbcMaterialRepository` save metadata, original BLOB, and extracted text together. `StudyPlannerCli` adds `upload`, `materials`, and `content`.

Rules: up to 10 files, 10 MiB each; unsupported, empty, corrupt, encrypted, or text-free files are rejected. No OCR is provided. Text is limited to 1,000,000 characters, PDF to 500 pages, and expanded DOCX to 20 MiB/1000 entries. API batches roll back on any failure; GUI files upload independently. Originals and text survive restarts.

F03: Document Summarization. AI-based. References: Stage 1 F03, UC-03, SD02 summary branch, and Task 4 F03. Select Summarize document; the agent summarizes source sections, synthesizes them

when needed, resolves selected source-excerpt IDs to verified quotations, and saves the result. View saved summary reuses it; regeneration explicitly requests a replacement. CLI: `summarize`, `summary`, `regenerate-summary`. Video pending.

Part	Implementation
GUI / API	`frontend/.../views/SummaryActions` runs background requests; `CourseApi.summarize/viewSummary` call `backend/.../resource/SummaryController`. Progress and safe errors remain visible while inference runs.
Agent / resource	`Course.generateStudyResource` calls `StudyAssistantFacade.summarize`. `ContentAgent.summarize` plans sections and validates responses from `AIEngine.generate`. `StudyResourceFactory.createResource` constructs `Summary`, extending `StudyResource`; `view` returns its text.
Storage	`SummaryService` verifies ownership, guards concurrent generation, and checks the source hash. `SummaryRepository/JdbcSummaryRepository` save the latest summary, evidence, model, and usage. Failed regeneration preserves the previous result.

F03 accepts 40–32,000 source characters. Generation has schema-constrained output, bounded calls/time, and one repair attempt for invalid results. Missing setup, provider failures, insufficient text, invalid output, and timeouts return errors without saving. Quotations must match the source; factual accuracy still needs human review.

F04: Concept Extraction. Implemented, AI-based. References: Stage 1 F04, UC-04, SD02 concept branch, Task 4 F04, and original UML. Select Extract Concepts for an uploaded document or uploaded section; a list shows important topic names, explanations, and supporting source quotations. View saved concepts and Regenerate concepts provide reuse and replacement. Video pending.

Part	Implementation and collaboration
GUI / API / CLI	`ConceptActions` runs background requests via `CourseApi.extractConcepts/viewConcepts`. `ConceptSetController` exposes GET/POST `/api/courses/{courseId}/materials/{materialId}/concepts`. CLI: `extract-concepts`, `concepts`, `regenerate-concepts`.
Agent / patterns	`Course.generateStudyResource` → `StudyAssistantFacade.extractConcepts` → `ContentAgent.extractConcepts` → singleton `AIEngine.generate`. The agent reads `LearningMaterial.getContent`, plans sections, extracts concepts, and consolidates multi-section results. `StudyResourceFactory.createResource` creates `ConceptSet`, the second concrete `StudyResource` product.
Validation / storage	Structured output requires 1–10 concepts with names, explanations, and valid source-excerpt IDs. Concept names may paraphrase the source; readable source sentences and long-sentence tails are available as evidence, with dense lists grouped into contiguous windows. Normalized duplicate names are merged; invalid output gets one repair attempt. `ConceptSetService` checks ownership, source hashes, and concurrent requests. `ConceptSetRepository/JdbcConceptSetRepository` atomically save the latest set in H2; failed regeneration preserves it.

F04 shares F03's 40–32,000-character, 14-call, 180-second, and provider limits. It operates on a selected uploaded material, including a separately uploaded section; course-section management and whole-course extraction are not included. Exact quotations do not guarantee explanation accuracy or semantic uniqueness.

F05: Question Answering Over Materials. Implemented, AI-based. References: Stage 1 F05, UC-05, SD02 question branch, Task 4 F05, and original UML. Open a course, enter a question in Course chat, and send it. The agent finds relevant course passages, asks Ollama for an answer, checks its source references, and saves the exchange. Expand a source to read its quotation or open the document text. Video pending.

Part	Implementation and key methods
GUI / API / CLI	`CourseChatPanel` shows progress, saved messages, wrapped answers, source quotations, and errors through `CourseApi.askQuestion/chatHistory`. `ChatController` exposes POST/GET `/api/courses/{courseId}/chat`. CLI: `ask` and `history`.
Domain / AI	`Course.askQuestion` → `CourseChat.askQuestion` → `StudyAssistantFacade.answerQuestion` → `ContentAgent.answerQuestion` → `AIEngine.generate`. `CourseContextRetriever.retrieve` ranks passages using words in the question and recent conversation. The model receives up to six matching passages. Follow-up questions can also use short excerpts from two recent exchanges; new topics start without earlier answers. F05 reuses Facade and Singleton.
Storage / validation	`ChatService` checks ownership, question length, and concurrent requests. `ChatRepository/JdbcChatRepository` save complete `ChatTurn` records in H2. `CourseChat.viewHistory` returns saved exchanges. Citations include material ID, filename, text offsets, and the original quotation. `ChatAnswerValidator.isUnfinished` detects dangling introductions and empty list items. History loads in pages of 50, with older messages available.

Questions allow 1–1000 characters. Chat searches up to 50 materials and 1,000,000 extracted characters per course; larger courses get an error. Invalid or visibly unfinished AI output gets one correction attempt. Introductions without the promised content and empty list items are not saved. Failed requests preserve the question in the GUI and leave history unchanged. Insufficient-context responses are saved without citations. Broad questions such as "Tell me about this course" use passages from across the uploaded materials. For specific questions with no matching passages, the system declines without a model call; the GUI labels this as Course search. Word matching can miss relevant passages, and valid source references do not guarantee correct answers.

F06: Personalized Study-Plan Generation; UC-11: Maintain Study Profile. Implemented, hybrid. References: Stage 1 F06, UC-06/UC-11, SD03, Task 4 F06, and original UML. Open Study plan and availability, save a profile, add busy times or quiz/exam dates, then choose plan dates and session count/duration. The AI recommends topics and activities; Java finds valid times and saves the plan and calendar sessions. Video pending.

Part	Implementation and key methods

GUI / CLI / API	`StudyPlanningPanel` edits availability, preferences, habits, and events, then displays priorities, sessions, and source quotations. `CourseApi` calls `PlanningController`. CLI adds `profile`, `save-profile`, `plan`, `generate-plan`, `events`, `add-event`, and `remove-event`.
Profile / orchestration	`StudyProfile.updateAvailability/updatePreferences` validate the shared weekly profile. `PlanningService` checks ownership, dates, material limits, and available time. `Course.generateStudyResource` calls `StudyAssistantFacade.generateStudyPlan`.
AI / scheduling / factory	`PlanningAgent.generateStudyPlan` uses source excerpts, profile data, upcoming dates, and optional assessment notes through the singleton `AIEngine.generate`. It validates topics, priorities, activities, and source IDs, with one repair attempt. `StudyScheduler.allocate` chooses all times. `StudyResourceFactory.createResource` creates `StudyPlan` and `StudySession`; `getSessions` exposes the schedule.
Storage / Observer	`PlanningRepository` stores profiles, latest plans, and events in H2. `PlanPersistence.save` attaches the `Calendar` observer and calls `StudyPlan.notifyObservers`. `Calendar.update` replaces the course's generated events. Plan and calendar changes commit together; failure preserves both previous records.

Weekly study minutes are shared across courses and must fit non-overlapping availability windows. Plans contain 1–30 sessions of 15–180 minutes over 1–56 days, within course dates and no earlier than today in the profile's time zone. Sessions avoid calendar conflicts, respect weekly budgets, and finish before the earliest upcoming quiz/exam in the selected period. Changing profile/events affects the next generated plan. Assessment results can be entered as notes; automatic quiz-result integration waits for F08/F09. Topic quality needs source review. Limits remain 50 materials / 1,000,000 extracted characters and two model calls.

Remaining Stage 1 scope:

ID	Feature	ID	Feature
F07	Flashcard Generation — pending	F08	Quiz Generation — pending
F09	Automatic Quiz Grading — pending	F10	Calendar — partial; full interface and Course/Quiz subjects pending

3. Design patterns and diagrams

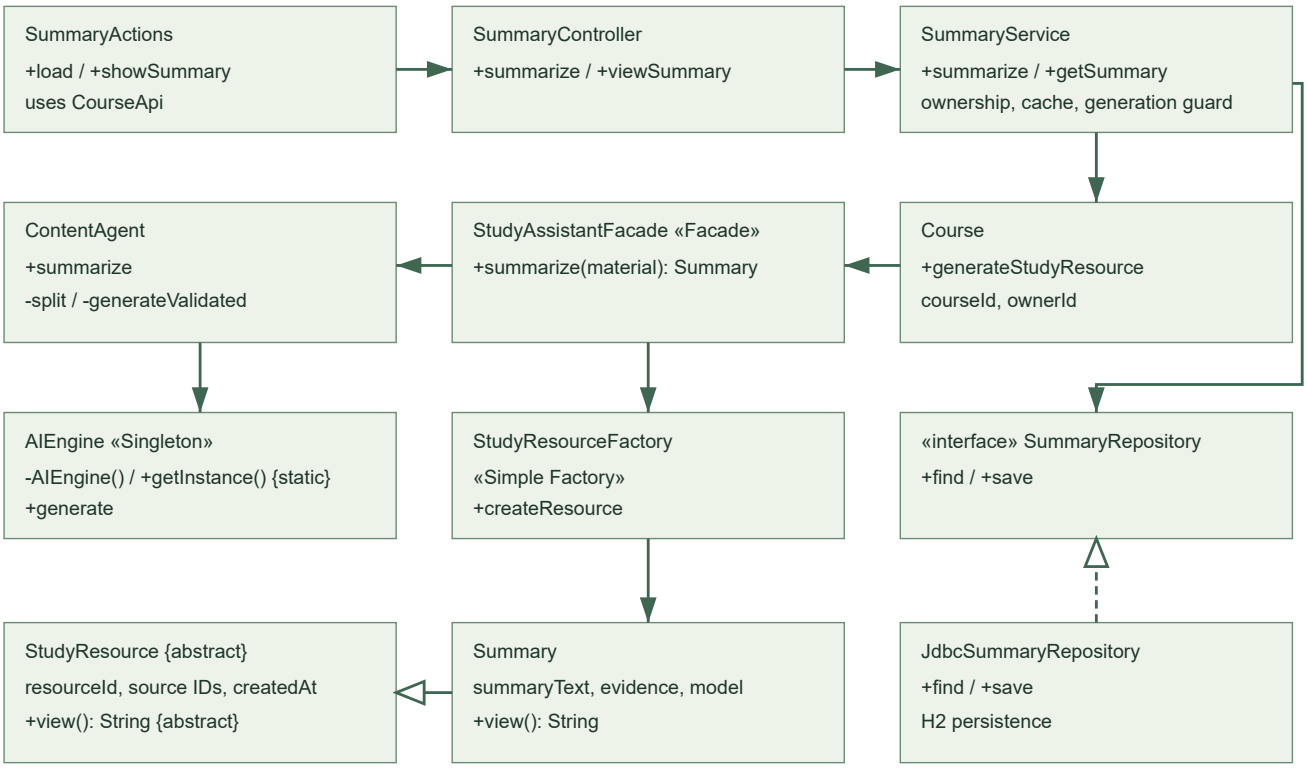
Stage 1 pattern	Implementation and purpose
Facade	`StudyAssistantFacade.summarize/extractConcepts/answerQuestion/generateStudyPlan` coordinates the agent and resource factory, hiding model orchestration and construction from course/UI callers.
Singleton	`AIEngine` has a private constructor and holder-based `getInstance`. Agents share one HTTP connection pool and concurrency limit; provider settings remain immutable per call.

Factory	`StudyResourceFactory.createResource` centralizes resource identity, timestamp, source association, and provenance. This simple factory constructs `Summary`, `ConceptSet`, and `StudyPlan` from validated agent data; other products remain pending.
Observer	`ScheduleSubject` defines attach/detach/notify; `StudyPlan` is the concrete subject. `ScheduleObserver` defines update; `Calendar` is the concrete observer. `notifyObservers` sends the changed plan to `Calendar.update`, keeping saved study events synchronized without StudyPlan containing calendar SQL. `PlanPersistence` makes both writes atomic. This follows the Stage 1 role assignment; the update method now receives the plan. Course/Quiz subjects remain pending.

Strategy remains pending. `CourseApiFactory` and Spring bean scopes are not counted as additional patterns.

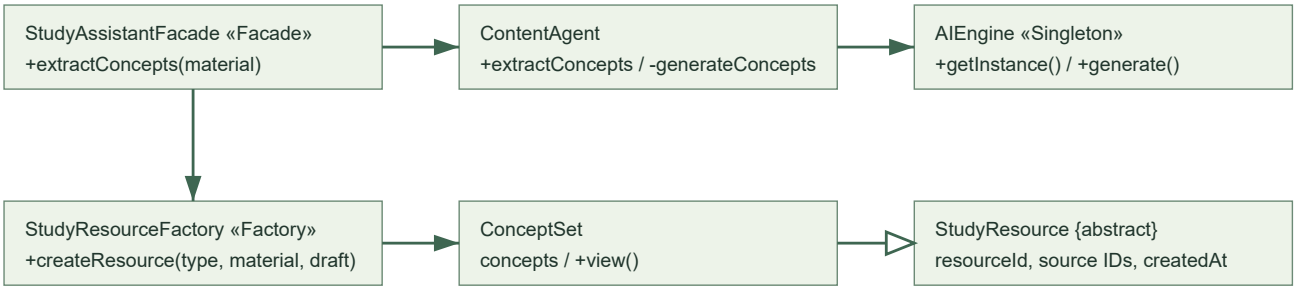
F01 adds a Repository abstraction: `CourseService` depends on `CourseRepository`, implemented by `JdbcCourseRepository`. This keeps SQL separate from application logic; it does not replace a planned pattern.

F03 UML class relationships; F01/F02 diagrams are linked below:

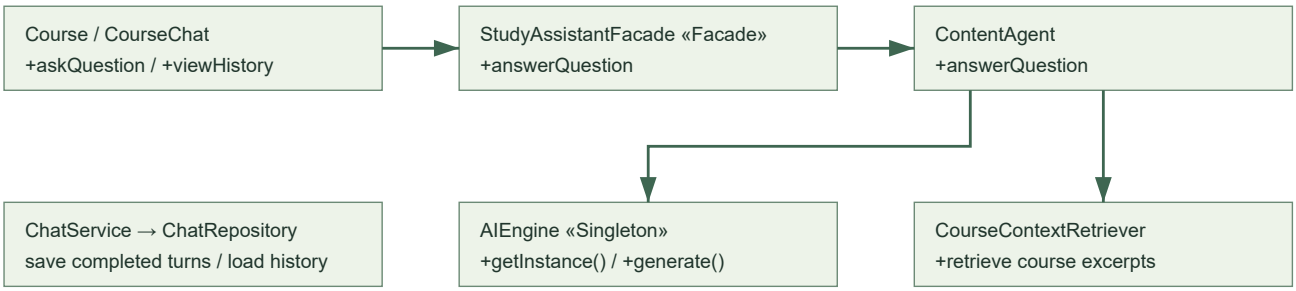


Updated UML/sequence diagrams: `documentation/diagrams/stage2-f01.md`, `stage2-f02.md`, `stage2-f03.md`, `stage2-f04.md`, `stage2-f05.md`, and `stage2-f06.md`. They show the implemented classes and pattern roles; original Stage 1 diagrams are preserved.

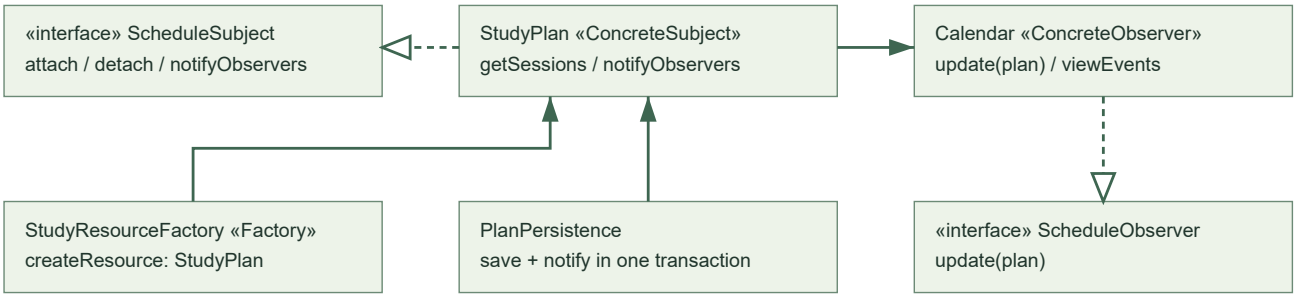
F04 adds a second product to the same pattern structure (full class and sequence details: `stage2-f04.md`):



F05 extends the same facade and singleton with course chat. Full class and sequence diagrams: `documentation/diagrams/stage2-f05.md`.



F06 adds StudyPlan as an Observer subject and a third factory product. Full class/sequence diagrams: `stage2-f06.md`.



4. Changes and rationale

Stage 1 design	Implementation change	Reason and impact
UI → CourseController → User → Course (F01/UC-01/SD01).	Added CourseApi, CourseService, CourseRepository, JDBC, and H2.	Connects the separate apps and saves courses across restarts. The creation sequence is updated.
User owns a collection of courses.	User creates courses; repository queries retrieve them by owner UUID derived from username.	Supports persistent ownership. Changing the username changes the owner identity; account persistence is future work.

Authenticated student required; authentication mechanism unspecified.	One configured account, Spring Security, HTTP Basic, and CSRF protection.	Enforces the precondition. Credentials stay in ignored local configuration. Remote deployment requires HTTPS.
Four course fields; detailed limits unspecified.	Required fields, text limits, trimming, and ordered dates.	Rejects invalid courses consistently. Editing/deletion remain pending.
GUI and CLI required.	Java CLI calls the same REST API as the GUI.	Reuses validation and storage. Future agent features still need CLI support.
F02/UC-02/SD01: CourseController → Course.addMaterial → LearningMaterial; file rules unspecified.	Added course navigation, MaterialService, parser, repository, text previews, and browser downloads of originals. PDF/DOCX/TXT, size and extraction limits apply.	Students can read extracted text or save the complete document, preserving formatting and images. Durable BLOB/CLOB storage keeps originals and text consistent. No AI or OCR is claimed.
F03/UC-03/SD02: Facade → ContentAgent → shared AIEngine → resource Factory → Summary.	Agent returns validated draft data for factory construction; section planning, synthesis, indexed source evidence, cache/source hashes, and explicit regeneration were added.	Handles longer documents and malformed model output with bounded work. SummaryService adds ownership and persistent memory. These additions affect the F03 sequence; original pattern roles are retained.
F04/UC-04/SD02: use the facade, agent, AI engine, and factory to create and display a ConceptSet. Duplicate concepts or invalid AI output cause an error.	Kept the original classes and patterns. Added processing for long documents, saved concepts, viewing saved results, and regeneration. Matching concept names are merged instead of causing an error. Invalid AI output gets one correction attempt before an error is shown.	Users can reuse saved concepts and process longer documents. The agent checks the AI response before the factory creates the ConceptSet. If generation fails, previously saved concepts are kept. These additions support the original F04 feature.
F05/UC-05/SD02: CourseChat asks the facade and agent for an answer from course materials and saves the conversation.	Kept the original classes and methods. Added a search for relevant passages, course overviews, source quotations, recent conversation context, history pages, and one correction attempt for invalid or unfinished AI output. Answers return text plus source information instead of a plain String.	Helps users check answers and ask follow-ups. ChatService and the repository save complete exchanges and enforce course ownership. Word matching may miss relevant material, and the model can still make mistakes. No new design pattern is claimed for F05.
F06/UC-06/UC-11/SD03: AI chooses study topics and activities, Java schedules sessions, and StudyPlan notifies Calendar.	Implemented that flow with saved weekly windows, a shared minute budget, time zones, blocking events, and latest-plan storage. Calendar.update receives the changed plan. The factory assembles sessions from Java-selected times. Course-wide plans carry citations instead of one material ID. Assessment notes are entered manually until quiz grading exists.	Makes availability and deadlines concrete and prevents conflicts across courses. The same plan/calendar transaction avoids partial updates. Facade, Singleton, Factory, and Observer keep their planned roles. Full F10 calendar views and automatic quiz-result input remain future work.

Stage 1 proposed ChatGPT, LangChain/RAG, and shared AIEngine.	Direct Java HTTP orchestration; holder-based singleton accepts provider settings per call. Local Ollama is selected.	Keeps the existing Java apps and enables local inference. F05 now retrieves course passages using word matching before generation. No LangChain, embeddings, or vector service is used.

No original feature or planned pattern has been removed or replaced.

5. AI tools and collaboration

Development tool: OpenAI Codex, a GPT-6-based coding agent; exact runtime model ID unavailable. Used for F01–F06 code, tests, diagrams, and reporting. **Application LLM:** local Ollama with qwen2.5:1.5b; no hosted API billing.

Interaction: the student requested F01/F02, a shorter report, then F03 with local Ollama and F04–F06 following the original UML. Codex implemented and checked the course, material, summary, concept, chat, and study-planning flows.

Tokens and cost: Codex totals remain unavailable; add platform usage/billing before submission. Successful summaries, concept sets, chat exchanges, and study plans retain provider-reported token counts; -1 means unavailable. Local compute cost is not measured. Human review remains required.

6. Development and verification

Seventy-four backend tests and the production frontend build pass. Coverage includes F01/F02 regression, F03 ownership/CSRF, section planning, output repair, quotation checks, cache/regeneration, timeouts, provider errors, and concurrent generation. Scripted-provider Chrome/CLI checks pass for summary progress, dialogs, saved results, regeneration, errors, mobile layout, and persistence. These fixtures test integration, not model quality. Real qwen2.5:1.5b verification also passed: CLI generation, browser viewing/regeneration, and a two-section document synthesized in three model calls. Source quotations were checked and the sample output reviewed. This verifies a working local integration, not general model accuracy.

Checks: `backend/mvnw.cmd clean verify`, frontend `mvnw.cmd package -Pproduction`, and `scripts/verify-f01.ps1`. Runtime checks use isolated test data and stop their own servers. A missing BCrypt encoding prefix was found and corrected during testing.

F02 runtime checks use `scripts/verify-f01.ps1 -IncludeF02`. Real-client checks caught a text-preview Accept-header mismatch and a parallel-upload CSRF session race. Correct response headers and serialized token/write requests fixed them.

Live testing exposed paraphrased quotations and missing JSON fields in the small model; constrained output and indexed evidence corrected these failures. F03 checks: `scripts/verify-f01.ps1 -IncludeF03`

uses a test-only provider; `scripts/verify-f03-live.ps1` uses real Ollama and a safe lecture sample. No production fake-response fallback exists.

F04 adds thirteen backend checks, including paraphrased names, full excerpt coverage, dense lists, repair feedback, ownership, cache/regeneration, and provider failures. `verify-f01.ps1 -IncludeF04` covers GUI/CLI and restart persistence; `verify-f03-live.ps1 -IncludeF04` passed real Ollama software-design/biology extraction, browser regeneration, and multi-section consolidation. A reported failure exposed literal-name matching against 12 sampled excerpts. Removing that rule and expanding coverage fixes false rejections while retaining source-ID checks. The small model can still produce broad topics, semantic duplicates, inaccurate explanations, or irrelevant citations; exact quotations establish provenance, not factual accuracy.

F05 adds eighteen backend checks for retrieval, ownership, citations, history, failures, overviews, and unfinished answers. GUI/CLI checks cover source viewing and persistence after restart. Live Ollama checks cover questions, follow-ups, unrelated topics, and overviews. Testing corrected a missing response schema, overly strict keyword checks, and earlier refusals affecting new topics. A reported answer stopped after "follow these steps:"; the agent now rejects dangling introductions and empty list items and requests one complete replacement. Tests verify that failed correction preserves history and complete short answers remain valid. Real PDF/CLI/browser verification (`verify-f05-completeness.ps1`) returned substantive guidance for the reported Stage 2 question. A separate layout regression reproduced long answers running beyond the right edge when custom wrapping was missing. Chat answers now enforce wrapping and width limits directly. Browser checks compare the full saved answer and its final sentence at 1920, 1280, and 390 pixels, including without the custom text class. These checks do not guarantee factual accuracy or detect every incomplete answer.

F06 adds eighteen backend checks for profiles, availability, weekly budgets across courses, busy times, deadlines, daylight-saving durations, ownership, provider errors, concurrency, and atomic Observer rollback. `verify-f01.ps1 -IncludeF06` passes GUI/CLI editing, regeneration, insufficient-time errors, mobile layout, and profile/plan/calendar persistence after restart with the provider stopped. Real Ollama (`verify-f06-live.ps1`) generated and regenerated a four-session biology plan around a work block and quiz deadline. Quotes were checked and recommendations reviewed. The model still needs human review; deterministic scheduling checks are independent of model quality.

Toronto reporting period	Minimum code commits	Observed local evidence
October 20–26, 2026	5	No qualifying local commits found; remote evidence needs checking.
October 27–November 2, 2026	5	a0db921: starter; 42853d5: F01; 8ab568b: F02. b26894b: F03. b0c6578: F04. Five local code commits observed; push status not verified.
November 3–9, 2026	5	No new local commits observed yet. F05/F06 are uncommitted.

Local credentials and database files are ignored by Git. `backend/config.local.properties.example` provides placeholders; setup instructions are in the README. F01 requires no API key.

Before November 10: complete F07–F10, Strategy and additional resource products, full-system diagrams, videos, cost records, and commit evidence. Check additional baseline instructions; only the supplied Stage 1 report and Stage 2 instructions were reviewed.

7. Demonstration index

Feature	Video status	Demonstration scope
F01 Course Creation	Recording and accessible link/timestamp pending.	Identify F01; show GUI and CLI creation, input errors, dashboard output, and persistence. Explain that F01 is deterministic. Keep credentials hidden.
F02 Upload Learning Materials	Recording and link/timestamp pending.	Show course navigation, GUI/CLI upload, saved files, preview, original downloads, invalid/type/size errors, and persistence. Identify deterministic extraction and pending AI processing.
F03 Document Summarization	Recording/link pending.	Show real Ollama GUI/CLI generation, source quotations, saved summaries, regeneration, and error cases. Explain section planning and output validation.
F04 Concept Extraction	Recording/link pending.	Show real GUI/CLI extraction, concept names/details and evidence, saved viewing, regeneration, input/provider errors, and section consolidation.
F05 Course Questions	Recording/link pending.	Show GUI/CLI questions, follow-ups, source viewing, unsupported questions, errors, and saved history after restart.
F06 Personalized Study Plans / UC-11	Recording/link pending.	Show profile editing, busy/quiz events, real AI recommendations, conflict-free sessions, calendar sync, regeneration errors, and saved data after restart through GUI/CLI.
F07–F09	Pending implementation.	Add labeled video segments.
F10 Calendar	Partial; video pending.	F06 demonstrates event inputs and synchronized study sessions. Complete the full calendar feature later.

Automated browser checks do not replace demonstration videos. Update this source and regenerate `stage2-report.pdf` as features are completed.